

The Bare Hand

*Pre-PIE *de- as a free morpheme, the compositional decomposition of
*-tero-, and the bimorphemic structure of *deh₃-*

Paper 3 of the *der-* project. Builds on Chavis & Claude (2026a) and (2026b).
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Abstract

The first two papers of this project argued that pre-PIE *de- “hand, open palm” underlies both the lexical cognate cluster Greek δῶρον / Hesychian δάρις / Celtic *durno- / Latvian dūfe (Paper 1) and the PIE contrastive suffix *-tero- together with the numeral *duo-/dui- “two” as derivational siblings on the same root (Paper 2). The present paper completes the framework’s verbal and particle branches by going one step deeper: *der- itself decomposes into two independently attested morphemes — a bare hand-element *de- and a locative/spatial extension *-r- — and the same bare *de- underlies the give-verb *deh₃- via a verbal extension *-h₃-.

The bare *de- is independently attested as the directional particle *de/do “to, toward” across the IE family: English *to*, Old Church Slavonic *do*, the Greek allative clitic -δε productive in Mycenaean Greek by 1400–1200 BCE, Celtiberian DV in the Bronze of Novallas (Beltrán Lloris et al. 2021; Prósper 2022: 19 n.45), and the Gaulish compound verb *de-uor-buet-id* (Lezoux; Lambert 2002). The *-r- extension is independently attested as a locative/spatial morpheme across three independent scholarly traditions (Lipp 2009; Bauhaus 2019; Kahl 2024).

The contrastive *-tero- of Paper 2 decomposes compositionally as *de- (directional, hand) + *-r- (locative) + *-o- (thematic) = “at the side where one points.” The same *de- underlies *deh₃- = *de- + *-h₃-: a verbal derivative of the hand-noun whose laryngeal is a derivational extension suffix. *der- (nominal) and *deh₃- (verbal) are siblings.

The new §4 develops the strongest single piece of discriminating evidence the framework has at any morphological stratum: the Greek root-aorist 2pl imperative paradigm shows a clean binary contrast between roots the framework analyzes as bimorphemic (*deh₃- → δότε, *dheh₁- → θέτε, *yeh₁-/seh₁- → ἔτε; bare-CV form preserved) and roots the framework does not analyze as bimorphemic (*steh₂- → στήτε, *gneh₃- → γνῶτε, *g^weh₂- → βῆτε; laryngeal-conditioned long vowel preserved). The asymmetry is descriptively well-established in the standard Greek-grammar tradition; the standard view describes it as paradigmatic accident, the framework predicts it from the bimorphemic decomposition.

The conditional probability is assessed at 35–45%, reflecting the new direct discriminating evidence in §4. The marginal probability for the framework as a whole, integrated across Papers 1–4, remains in the 1–4% range.

1. Introduction

The four-paper project rests on a single substantive proposal: that pre-PIE had a body-part etymon meaning “hand, open palm,” and that this etymon underlies a network of lexical, morphological, and verbal reflexes in Proto-Indo-European whose unity the standard handbooks have not recognized. Paper 1 (Chavis & Claude 2026a) makes the case for the lexical branch: the cognate cluster Greek δῶρον / Hesychian δάρις / Celtic **durno-* / Latvian *dūfe* reflects two derivational siblings on the same bare hand-element **de-*. Paper 2 (Chavis & Claude 2026b) extends the case into morphology: the PIE contrastive suffix **-tero-* descends from the same etymon via the grammaticalization pathway HAND > SIDE > BINARY-SPATIAL-CONTRASTIVE, and the numeral **duo-/dui-* “two” is a parallel derivational outcome. Paper 4 (Chavis & Claude 2026d) carries the framework into Anatolian, deriving Hittite *dā-* “to take” from the directive member of an early-PIE paradigmatic pair on the same nominal root.

The present paper goes one step deeper. **der-* itself is bimorphemic: a bare hand-element **de-* plus a locative/spatial extension **-r-*. The bare **de-* has independent attestation as the directional particle **de/do* “to, toward” preserved across the IE family. The **-r-* has independent attestation as a locative morpheme (Lipp 2009; Bauhaus 2019; Kahl 2024). The give-verb **deh₃-* itself decomposes parallel to the body-part noun: bare **de-* + verbal extension **-h₃-*, with the laryngeal as a derivational suffix rather than an unanalyzable root consonant. The new §4 develops the discriminating evidence for this bimorphemic decomposition through the Greek root-aorist 2pl imperative paradigm.

The paper is organized as follows. §2 documents the bare **de-* as a directional particle attested across the IE family, with particular attention to Mycenaean -δε and Celtiberian DV. §3 documents the locative status of **-r-* across three independent published treatments. §4 presents the Greek root-aorist 2pl imperative contrast as direct discriminating evidence for the bimorphemic decomposition of **deh₃-*. §5 develops the compositional decomposition of **-tero-*. §6 develops the parallel decomposition of **deh₃-* as **de-* + **-h₃-*. §7 places the framework’s typological predictions in the cross-linguistic record. §§8–11 close with future research, remaining problems, the conclusion, and the probability assessment.

2. The bare **de-* attested in PIE

§2.1. The directional particle **de/do* across the family

PIE **de/do* “to, toward” is a well-attested local/temporal particle. Its reflexes include Greek $\delta\acute{\epsilon}$ (directional/sequential origins, later grammaticalized as connective), English *to* (with German *zu*, Dutch *te*) from Proto-Germanic **tō*, Old Church Slavonic *do* “until, up to, toward,” and Celtiberian DV in the Bronze of Novallas (Z.02.01; Beltrán Lloris et al. 2021; reported by Prósper 2022: 19 n.45). Dunkel’s *Lexikon der indogermanischen Partikeln* (2014) treats **de* and **do* as ablaut variants of a single directional particle across the family.

The SW Palaeohispanic corpus (approximately 90–100 Iron Age inscriptions from the southwestern Iberian Peninsula, with the most securely dated example from Medellín assigned to 650–625 BCE) contains a systematic preverb that Koch (2019) analyzes as the directional element $|\text{de}\cdot|$. The compound verbs **tee-baare*, **tee-robaare*, and **teee-baarentii* are reconstructed as $|\text{de}\cdot\text{bāre}|$ “has carried away,” $|\text{de}\cdot\text{ro}\cdot\text{bāre}|$ (perfectivized), and $|\text{dē}\cdot(\text{e}(\text{n}))\cdot\text{bārenti}|$ “they have now carried away,” all < Proto-Celtic compound **dē* + $\sqrt{\text{berd-}}$ “carry,” cognate with Latin *dēferō* and Welsh *diferaf* (Koch 2019: 91–93). The parallel Gaulish compound *de-uor-buet-id* (Lezoux; Lambert 2002, RIG II L-66) independently attests a **de*-preverb in Transalpine Celtic. The Celtic classification of the SW corpus is contested; the SW Celtic evidence is registered conditionally.

Crucially, the Novallas bronze co-attests Paper 1’s body-part noun branch (**DERNV* = **de-r-nā-*, Stifter 2022) and the directional particle branch (DV = **de/do*) in a single Celtiberian inscription. This convergence is not visible in the existing literature; it becomes visible only through the framework’s lens.

The standard treatments list **de/do* as a primitive PIE particle with no further decomposition. The framework proposes the missing etymology: **de/do* is the bare descendant of pre-PIE **de-* “hand,” grammaticalized along the pathway HAND > REACH/EXTEND > DIRECTIONAL. The extended hand points; the word for “hand” is reanalyzed as a directional marker. The pathway is typologically well-supported: Heine & Kuteva (2002) document HAND > LOCATIVE/GENITIVE/POSSESSIVE and related developments across Niger-Congo, Austronesian, Sino-Tibetan, and Mayan.

§2.2. The Greek allative $-\delta\epsilon$ and its Mycenaean antiquity

The strongest evidence for the framework’s chronological claim comes from Greek, where the bare **de-* element is preserved as the productive allative clitic $-\delta\epsilon$. The clitic marks motion toward: $\omicron\iota\kappa\alpha\delta\epsilon$ “homeward” (from $\omicron\iota\kappa\omicron\varsigma$ “house”), $\Lambda\theta\acute{\eta}\nu\alpha\zeta\epsilon$ “to Athens” (with sandhi), $\pi\acute{o}\lambda\epsilon\mu\acute{o}\nu\delta\epsilon$ “to war” (*Iliad*). The function is unambiguously directional/allative — motion toward a destination, with no stative or ablative meaning.

Linear B tablets from approximately 1400–1200 BCE preserve *-de* as a fully productive directional clitic on toponyms, with dozens of attested forms catalogued by Lejeune (1961): *te-qa-de* /Thēg^wasde/ “to Thebes,” *a-mi-ni-so-de* “to Amnisos,” *da-da-re-jo-de* “to the Daidaleion,” among others (van Hes 2021). Three features of the Mycenaean attestation matter for the framework. *First, productivity*: Mycenaean *-de* is not a closed set of frozen relics but a productive clitic applying freely across the corpus. *Second, depth of grammaticalization*: productive grammaticalization by 1400 BCE implies that the underlying HAND > DIRECTIONAL development was substantially complete well before the proto-Greek period, plausibly extending back into PIE itself. *Third, semantic profile*: Mycenaean *-de* is narrowly directional/allative — motion toward, not stative location, not source — exactly the profile expected of a morpheme grammaticalized from “where the hand points/extends.”

§2.3. Competing accounts: the χαράι question

Greek χαράι “on/to the ground” has figured in debates about whether to reconstruct an allative case for PIE. Melchert (2017) reads it as a fossilized allative **ĝ^hōm-eh₂* and argues from it that the Anatolian allative in *-a* descends from PIE **-eh₂*. Beekes (2010) and Frisk (1960) treat the form as a petrified locative. We do not settle the question here; we note that it is genuinely open. Paper 4 §4 returns to the Anatolian allative *-a* and its relationship to the directional element in Hittite *dā-*.

§2.4. The Celtiberian directional prefix **dū-* in institutional vocabulary

The same Celtiberian **dū-* directional element appears as a verbal prefix in the institutional hospitium vocabulary. Prósper (2022: 13) derives Celtiberian *tureita/dureita* as **dū-rextā* “issued, directed” — where **dū-* is the directional prefix and **reχ-* is from PIE **reg-* “to direct, rule.” The preterite *tures* < **dū-reχ-s-t* “issued” (3sg past) is independently attested at Torrijo del Campo (Teruel) and at K.0.7 (Gortono; Prósper 2022: 19), and the participial form DVRETA SALDANICA appears in peripheral Hispano-Celtic (León).

These forms appear in and on hand-shaped hospitium tesserae (*téseras hospitales*) cast as clasped right hands (*dextrarum iunctio*), specifically on K.23.2 where *taruotureska tureita* accompanies the ethnic adjective for the city of *Tarvodurum*. The directional element Paper 3 traces to the pre-PIE hand-root surfaces in Celtiberian both as a standalone allative preposition (DV, Novallas; §2.1) and as the productive verbal prefix of the institutional vocabulary for right-hand pledge ceremonies. The semantic context — physical hand-extension formalized in legal agreements, on bronzes cast as clasped hands — is the domain from which the grammaticalization pathway HAND > DIRECTIONAL originates.

3. PIE **-r-* as a locative/spatial element

The locative status of **-r-* is the decompositional argument’s second morpheme and its most critical empirical claim. Three independent published treatments from three different scholarly traditions converge on the same conclusion.

§3.1. Lipp (2009): athematic locatives in **-er* and **-én*

Lipp’s *Die indogermanischen und einzelsprachlichen Palatale im Indoiranischen* (2009), in reconstructing the PIE heteroclititic r/n-stem paradigm, argues that the locative singular forms of acrostatic heteroclitite neuters are built from root nouns combined with an adverbial particle, yielding parallel locative endings **-er* and **-én*. On Lipp’s analysis, **-r* and **-én* are locative/spatial morphemes meaning “at/in the place of (X).” The spatial semantics is primary; the body-part association is downstream.

§3.2. Bauhaus (2019): **-r* as a true PIE locative case ending

Bauhaus (2019) advances a stronger version of the same conclusion. Where Lipp treated **-r* as an adverbial element, Bauhaus argues that **-r* is a true PIE locative case ending — “a real case ending, not an ‘Adverbialendung’,” in his words. He surveys pronominal and nominal formations including **k^wor* (Gothic *þor* “where”), **tor* (Gothic *þar* “there”), and nominal locatives such as **ǵ^hǵm-ér* “on the earth.” Bauhaus’s treatment is the strongest published case for **-r* as a productive locative case ending in early PIE.

§3.3. Kahl (2024): independent confirmation from degree morphology

Kahl (2024), *Diachronic Studies in Indo-European Degree Morphology* (Harvard PhD dissertation under Jeremy Rau), reaches the same locative conclusion through a separate scholarly tradition. Kahl’s abstract states explicitly: “the origin of the ‘secondary comparative’ of Indo-Iranian and Greek is traced to an archaic locative morpheme **-er*.” Three independent published treatments from three scholarly traditions converge on the locative **-r-*.

§3.4. The locative-directive paradigmatic pair

Bauhaus’s most structurally significant contribution for the framework is his reconstruction of a paradigmatic opposition between locative **-r* and directive **-o* attached to the same nominal roots. The pairs he reconstructs include **h₁upér* “on top” vs. **h₁upó* “to the top,” **k(e)nter* vs. **knto*, and **ndher* vs. **ndho*. This paradigmatic pair is the structural template the framework reconstructs across Papers 3 and 4: locative **de-r-* (body-part noun) versus directive **de + a* (Paper 4’s analysis of Hittite *dā-* “to take”). What was previously a framework-internal claim now has explicit external precedent in published reconstruction.

4. The Greek aorist 2pl imperative contrast: direct evidence for bimorphemic **deh₃-*

The bimorphemic decomposition of **deh₃-* as bare **de-* + **-h₃-* predicts a specific empirical asymmetry in the inherited Greek paradigm. Roots the framework analyzes as bimorphemic (a bare CV-stem plus a laryngeal extension) should preserve the bare pre-extension form in conservative paradigm slots; roots the framework does not analyze as bimorphemic (where the laryngeal is part of the unanalyzable root) should show the laryngeal's full effect throughout. The Greek root-aorist 2pl imperative paradigm is exactly such a conservative slot, and it shows the asymmetry the framework predicts. This constitutes the most direct discriminating evidence the framework has at any morphological stratum.

§4.1. The contrast: Class A short, Class B long

The Greek contrast is descriptively well-established in the standard Greek-grammar tradition (Goodell 1902 §368; Smyth §§682–691; Sihler 1995 §502). Class A roots — those the framework analyzes as bimorphemic — show the bare-CV form in 2pl. Class B roots — those the framework does not analyze as bimorphemic — show the laryngeal-conditioned long vowel.

PIE root	Framework analysis	Greek 2sg aor. impv.	Greek 2pl aor. impv.	Vowel length
<i>*deh₃-</i> “give”	bimorphemic: <i>*de-</i> + <i>*-h₃-</i>	δόϛ	δότε	short <i>o</i>
<i>*dheh₁-</i> “put”	bimorphemic: <i>*dhe-</i> + <i>*-h₁-</i>	θέϛ	θέτε	short <i>e</i>
<i>*yeh₁-/seh₁-</i> “let go”	bimorphemic	έϛ	έτε	short <i>e</i>
<i>*steh₂-</i> “stand”	monomorphemic root	στῆθι	στῆτε	long <i>η</i>
<i>*ǵneǵh₃-</i> “know”	monomorphemic root	γνῶθι	γνῶτε	long <i>ω</i>
<i>*g^weh₂-</i> “go”	monomorphemic root	βῆθι	βῆτε	long <i>η</i>

The pattern is uniform. The three Class A roots preserve a bare-CV form in 2pl, with no laryngeal effect on vowel length. The three Class B roots preserve the laryngeal-conditioned long vowel throughout the paradigm.

§4.2. The standard view's account and what it leaves unmotivated

The standard view's account of the asymmetry is that Class B roots underwent Greek-internal paradigm leveling that extended the long vowel from singular into plural, while Class A roots preserved the original ablaut alternation with zero-grade in plural. The asymmetry is descriptively real but structurally unmotivated on the standard view: leveling happened to one set of verbs but not another, for no stated reason. The standard treatments offer no answer to the question of why Class A and Class B should behave differently under leveling pressure.

§4.3. The framework’s account: bimorphemic anchoring

The framework’s account is structural. Class A roots have bimorphemic structure; their 2pl imperatives δότε / θέτε / ἔτε preserve the pre-extension bare CV-stem because that stem was independently anchored in pre-PIE — for **deh₃*- specifically, by **de*-’s status as a productive directional particle (§2 above), as a body-part noun (Paper 1), and as the source of the contrastive suffix **-tero-* (Paper 2). The bare **de-* of pre-PIE was not merely the zero-grade form of the verbal root; it was an independently active morpheme with substantial polyfunctional weight. When Greek inherited the paradigm, the bare-CV weak-stem form had multiple anchors holding it in place against leveling pressure.

Class B roots have no pre-extension form to anchor. **steh₂-* “stand,” **ǵneḥ₃-* “know,” **g^weh₂-* “go” — none of these admits a bimorphemic decomposition into a bare CV-stem plus a laryngeal extension that is independently attested. When paradigm leveling pressure operated on the inherited paradigm, the long-vowel singular stem extended into the plural without resistance. The framework’s bimorphemic decomposition therefore predicts the asymmetry: it predicts that Class A roots, with independently anchored pre-extension forms, should resist leveling in conservative paradigm slots, while Class B roots, without such anchoring, should level. This is exactly the pattern the data shows.

§4.4. The Sanskrit complication and the principle that emerges

A complication should be acknowledged honestly. The Greek contrast is Greek-internal and does not extend straightforwardly to Sanskrit. Sanskrit root-aorist imperative 2pl forms are leveled to long-vowel throughout the paradigm: *√dā* → *dāta*, *√dhā* → *dhātana*, *√sthā* → *sthāta*. Sanskrit leveled the long vowel across the paradigm in both Class A and Class B verbs. Where Sanskrit does preserve the bare-CV pattern is in the *present* paradigm: *dadhāti* → *datta* (2pl impv. of the reduplicated present, with cluster-simplified geminate), parallel to Greek δίδωμι → δίδετε. The Sanskrit witness for the framework is therefore in the present paradigm; the Greek aorist contrast is the cleanest single piece of discriminating evidence available across IE.

The principle that emerges is **weak-stem conservatism in athematic paradigms for bimorphemic roots specifically**: in conservative slots where the weak-stem ablaut conditions apply, bimorphemic roots preserve the bare pre-extension form because that form was independently anchored at pre-PIE depth, while monomorphemic roots show the laryngeal’s full effect because there is no pre-extension form to preserve.

§4.5. Control roots and the discriminating power of the test

The control for the test deserves brief discussion. **steh₂-* has a cluster onset (**st-*), raising the possibility that the long-vowel preservation in στήτε is conditioned by

the onset cluster rather than by the laryngeal’s morphological status. The cleaner controls are **ġneġh₃-* “know” (single onset **ġn-*, gives γνῶτε with long ω) and **g^weh₂-* “go” (single onset **g^w-*, gives βῆτε with long η). Both are single-onset CVH roots that the framework does NOT analyze as bimorphemic, and both show the laryngeal’s full effect in 2pl. The contrast therefore tracks the framework’s bimorphemic/monomorphemic distinction directly, not an artifact of onset shape.

What the test does and does not show: the test shows that the inherited Greek paradigm preserves a class-by-class asymmetry that aligns with the framework’s bimorphemic decomposition and is unmotivated on the standard monomorphemic view. The test does not show that the standard view is incorrect — the standard view’s description of the morphological history is accurate and consistent with the framework’s account. The framework’s advantage is etymological: it explains why Class A and Class B should behave differently, where the standard view leaves the difference as an unmotivated paradigmatic fact.

5. The compositional decomposition of **-tero-*

§5.1. The composition is morphologically transparent

With both pieces in place — bare **de-* (directional, hand) and **-r-* (locative) — the compositional decomposition of **-tero-* follows directly. The suffix decomposes as **de-* + **-r-* + **-o-* (thematic vowel). Each morpheme is independently attested with the right phonology and the right semantics. The semantics compose transparently: **de-* is “hand,” grammaticalized to “where the hand points”; **-r-* is “at the place of.” The combined **der-* is “at the place where the hand points/extends” — the side toward which one is oriented.

§5.2. The bilateral contrast: a structural consequence of the composition

Bare **de-* is an absolute directional: “toward there,” pointing at a target with no implied alternative. Greek οἰκαδε “homeward” does not presuppose a contrasted alternative direction. The locative **-r-* changes this. It anchors the directional to the body as frame of reference — and bodies have inherent bilateral symmetry. The bilateral character of **-tero-* is therefore a structural consequence of what **de-r-o-* composes to mean, rather than a grammaticalized accident of the suffix’s history. Paper 2 §4 develops a stronger reading through the morphological-identity argument: the bilateral character is a morphological inheritance from the **-u-* element of pairedness/twoness operating on the same root, with the inherited compound **ui-tero-* (Vedic *vítarám*, Avestan *vítara-*, OCS *vŭtoru* “the second”; Dunkel 2014) as the surface witness.

§5.3. Attestation in the Celtic derivational system

The compositional architecture the framework reconstructs — particle or directional element + derivational suffix → spatial or relational noun/adjective — is not confined to the reconstructed prehistory of PIE. Eska (2004) documents a productive system of preposition-derived nominals in Celtic that instantiates exactly this pattern in a directly attested IE branch.

The core system includes: *aitire* (**enter* + *-*īā* “between-suretyship”), *inne* (**in* + *-*īā* “quality, in-ness”), *echtrae* (**exter* + *-*īā* “out-expedition”), *aire* (**ari* + *-*īā* “noblesse, before-excess”), *íarmai* (**eφiro* + *-*īā* “remnant, after-lack”), and *uxsedia* (**uxse* + *-(*V*)*dio*- “total, over-things”). Lambert (1995) extends the paradigm with Gaulish derivatives *ambio*- “around,” *ario*- “before,” and *cantio*- “with.” The Celtic system shows that this morphological operation — directional particle + derivational suffix → spatial-relational adjective — was productive in at least one IE branch at the earliest stages of attestation. The framework is not proposing an unusual morphological type; it is proposing a PIE-stage instance of a type Celtic preserves.

§5.4. What the decomposition does for the framework

The compositional decomposition is one of the strongest single pieces of evidence for the four-paper framework as a whole. If the framework is wrong, then **de*- (the directional particle), *-*r*- (the locative element, supported by three independent published treatments), and *-*tero*- (the spatial-direction adjective) all happen to share phonology and semantics for the compositional reading by coincidence. The coincidence required is substantial: three independent matches converging on the predicted composition demand explanation. The framework provides one; the standard view does not.

6. **deh*₃- “give” as **de*- + *-*h*₃-

The decompositional logic of §5 extends to the give-verb. PIE **deh*₃- “give” is reflected in Sanskrit *dádāti*, Greek δίδωμι, Latin *dō/dare*, Old Church Slavonic *dati*, Lithuanian *dúōti*. The framework reads **deh*₃- as **de*- (the bare hand-element of §2) + *-*h*₃- (the laryngeal extension), where *-*h*₃- converts the bare hand-noun into a verbal stem. The semantics compose transparently: hand + (verbal extension) = the act of handing over = to give.

The nominal/verbal distinction resolves what earlier versions of Paper 1 treated as a cost. **der*- (= **de*- + *-*r*-) is a noun; δίδωμι and its cognates are built on **deh*₃- (= **de*- + *-*h*₃-), the verbal derivative. The /i/-reduplication of δίδωμι belongs to **deh*₃-’s morphology and is not a problem for the framework.

The framework’s claim about **deh*₃- is morphological, not phonological. We do not propose that **h*₃ is absent from the verbal stem in post-Anatolian PIE — the Anatolian evidence for its effects (Kloekhorst 2013; Norbruis 2021) is real, and we

accept it for the post-Anatolian period. We propose that **h₃* was not always there root-internally as a constituent of an unanalyzable primitive. It was added to **de-* as a derivational extension suffix at a post-Anatolian stage, converting a directional noun into a verbal stem. Paper 4 develops this chronological claim through the analysis of Hittite *dā-* as preserving a pre-Anatolian **de + a* univerbation that does not contain the laryngeal extension. The §4 evidence above is the strongest single piece of paradigmatic support: the Greek root-aorist 2pl imperative paradigm preserves bare δότε (and θέτε, ἔτε) without the laryngeal effect, while the comparable monomorphemic roots preserve στήτε (and γνῶτε, βῆτε) with the laryngeal effect.

7. Cross-cultural typology

The framework's typological predictions for this paper are HAND > DIRECTIONAL (underlying the bare **de-* of §2) and HAND > GIVE (underlying **deh₃-* of §6). Both are well-attested in the cross-linguistic record. HAND > DIRECTIONAL is a natural specialization of the documented HAND > LOCATIVE pathway (Heine & Kuteva 2002). HAND > GIVE is documented as one of the most natural verbal grammaticalizations from a body-part noun, attested in Niger-Congo, Austronesian, and Sino-Tibetan languages (Heine & Kuteva 2002: 167-168).

8. Future research

**dher-* “hold, maintain” and Sanskrit *dharma*. The verbal root **dher-* is reflected in Sanskrit *dhārayāti* “holds, maintains,” Old Iranian *dāraya-* “holds” (preserved in the royal name Darius: “he who holds firm”), Greek θρόνος “throne,” and Sanskrit *dharma* “cosmic order, law, righteousness.” The framework reads **dher-* as a further derivative of **de-*: the hand that holds rather than the hand that gives.

PIE **terh₂-* “to cross over.” This verbal root gives Latin *trāns* “across,” English *through*, Sanskrit *tāraṭi* “passes, overcomes.” Whether **terh₂-* is etymologically related to **der-* through a chain HAND > REACH-ACROSS > CROSS-OVER is flagged as promising terrain.

The Celtiberian institutional register. The Novallas bronze co-attestation of **DERNV* (Paper 1) and DV (Paper 3), together with the directional prefix **dū-* in the hospitium verb *tureita/tures/dures* (§2.4), means that Celtiberian hospitium epigraphy contains evidence for three separate branches of the bare hand-element in a single archaeological and cultural register: the body-part noun, the directional particle, and the give-verb. A full treatment is deferred to future work.

The §4 contrast and broader paradigm-conservatism testing. Whether the same bimorphemic/monomorphemic asymmetry can be constructed for other paradigm

slots — Greek perfect imperatives, Sanskrit present subjunctives, Tocharian preterites — is a natural extension. If the framework is right, the asymmetry should be recoverable in other conservative weak-stem positions across the family.

9. Remaining problems

Four problems deserve explicit statement; none is fatal.

First, the standard etymology of Greek -δε does not unanimously trace it to PIE **de/do*; alternative reconstructions exist. The χαράι case is genuinely contested (§2.3).

Second, the locative status of **-r-* now rests on three independent published treatments that converge on the same conclusion. This is the framework's strongest morphological support, and the key development since Pokorny.

Third, the property suffix **-ro-* (in **rudh-ró-* “red”) and the instrument suffix **-tro-/trom* contain **-r-* but lack obvious locative semantics. The framework needs an account of the semantic divergence between the locative and the property/instrument uses.

Fourth, the proposed function of the laryngeal extension **-h₃-* in **deh₃-* is reconstructed rather than directly motivated by §4 alone. The §4 evidence supports the bimorphemic decomposition but does not by itself specify what the laryngeal extension is doing semantically. On the Novallas DV evidence specifically: Prósper (2022: 19 n.45) provides only a parenthetical cross-reference; the primary edition Beltrán Lloris et al. (2021) is the authority for the DV reading.

10. Conclusion

**der-* itself is bimorphemic: bare **de-* — the atomic hand-element, independently attested as the PIE directional particle **de/do* (most directly in the Mycenaean allative clitic -δε productive by 1400 BCE, in Celtiberian DV at Novallas, and conditionally in SW Celtic **tee* and Gaulish *de-uor-buet-id*) — plus **-r-*, the locative/spatial morpheme supported by three independent published treatments (Lipp 2009; Bauhaus 2019; Kahl 2024).

The bimorphemic decomposition of **deh₃-* — bare **de-* + verbal extension **-h₃-* — has direct discriminating evidence in the Greek root-aorist 2pl imperative paradigm (§4). Class A roots that the framework analyzes as bimorphemic preserve a bare-CV form (δότε, θέτε, ἔτε); Class B roots that the framework does not analyze as bimorphemic preserve the laryngeal-conditioned long vowel (στῆτε, γνῶτε, βῆτε). The asymmetry is descriptively well-established in the standard Greek-grammar tradition; the standard view describes it as paradigmatic

accident; the framework predicts it from the bimorphemic decomposition.

The framework makes a specific etymological commitment at each step where the standard view leaves the question open. **de/do* is either a primitive PIE particle of unknown origin (standard) or the bare descendant of a pre-PIE hand-noun (framework). **-r-* is either a heteroclitic alternant (standard) or a single locative case ending (framework, following Lipp, Bauhaus, Kahl). **-tero-* is either a primitive suffix of unknown etymological origin (standard) or **de-* + **-r-* + **-o-* (framework). **deh₃-* is either a primary verbal root (standard) or **de-* + **-h₃-* (framework, with §4 paradigmatic evidence). At each step the framework provides a unified motivation where the standard view leaves a gap. The IE etymology of the directional particle, the contrastive suffix, and the give-verb is on the table.

11. Probability estimate

Conditional on both Papers 1 and 2 being correct, the probability that the substantive thesis of this paper is correct in its basic outline is approximately 35–45%. Two factors motivate this range.

First, the Greek root-aorist 2pl imperative contrast (§4) is the strongest single piece of discriminating evidence the framework has at any morphological stratum. Where every other argument for bimorphemic **deh₃-* is structural, the §4 evidence is direct paradigmatic data: a class-by-class asymmetry in the inherited Greek imperative paradigm that aligns with the framework’s bimorphemic decomposition and is unmotivated on the standard monomorphemic view. The data is descriptively well-established in the standard Greek-grammar tradition; what the framework adds is a structural explanation.

Second, the discriminating power of the test is asymmetric. The test does not show that the standard view is incorrect — the standard view’s description of the morphological history is accurate and consistent with the framework’s account. The framework’s advantage is etymological: it explains why Class A and Class B should behave differently, where the standard view leaves the difference as an unmotivated paradigmatic fact. What would push the estimate toward 45%: independent paradigm-slot tests recovering the bimorphemic/monomorphemic asymmetry in other conservative weak-stem positions across the family. What would push toward 35%: a competing analysis that derives the Greek root-aorist 2pl asymmetry from a phonological or analogical mechanism not requiring bimorphemic decomposition.

The marginal probability — unconditioned, integrating over the probability that Papers 1 and 2 are also correct — is approximately: 30–40% (Paper 1) × 35–45% (Paper 2) × 35–45% (Paper 3) ≈ 3.7–8.1%. Much of the evidence supports multiple strata simultaneously rather than each stratum independently; the honest marginal range for the framework as a whole, integrated across Papers 1–4, is in

the 1–4% range. We continue to regard the central claim as more likely wrong than right, and to regard it as interesting enough, and the evidential structure dense enough, that publication for examination by other scholars is warranted.

12. Authorship and contributions

Chavis’s contributions. The hypothesis that the directional particle **de/do* — specifically as preserved in Greek $-\delta\epsilon$ — descends from a free-standing pre-PIE hand-element originates with Chavis. The decision to pursue the bimorphemic decomposition of **der-* itself is Chavis’s. The decision to add §4 — making the Greek root-aorist 2pl contrast the load-bearing discriminating evidence for bimorphemic **deh₃-* — is Chavis’s, made after the planning conversation for this version identified the contrast as the strongest single piece of paradigmatic evidence available to the framework but previously without a settled home across the project’s papers. Chavis directed the Mycenaean $-\delta\epsilon$ argument (§2.2) and the Celtiberian institutional vocabulary section (§2.4), and made the decision to frame the Novallas convergence of **DERNV* and DV as visible only through the framework’s lens. All fundamental decisions about scope, framing, and direction rest with Chavis, who supplies all intellectual continuity across sessions through systematic transcript management.

Claude’s contributions. Claude drafted the §4 sequence in five subsections and integrated it between §3 (the locative **-r-*) and §5 (the compositional decomposition of **-tero-*), so that the discriminating evidence supports the bimorphemic decomposition of **deh₃-* before the parallel decomposition of **-tero-* is developed. Claude tightened §2.2 around the three-part argument (productivity, depth of grammaticalization, semantic profile) from prior threads. Claude located and read Lejeune (1961) and van Hes (2021) on the Mycenaean $-\delta\epsilon$ attestations; Kloekhorst (2013) on the Hittite TA/DA phonetic distinction; Norbruis (2021) on the semantic chronology of **deh₃-*; Koch (2019) on the SW Celtic **tee* preverb; Lambert (1995, 2002) on the Gaulish compound verb and derivational paradigm; Beltrán Lloris et al. (2020, 2021) on the Novallas bronze and the hospitium tessera class; Eska (2004) on the Celtic preposition-derived nominals; Goodell (1902), Smyth, and Sihler (1995) on the Greek root-aorist paradigm; and Bauhaus (2019) on the locative-directive paradigmatic pairs. Claude proposed the structuring of the control-root argument of §4.5 and the bimorphemic-anchoring framing of §4.3. Claude drafted all prose sections from Chavis’s structural and substantive direction. The probability estimate in §11 is Claude’s own. Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management. The collaboration is genuine. The argument was built together. Its strengths and weaknesses belong to both authors, with the responsibility resting with Chavis.

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